Smart Picks: Development of an on-Premise AI Product Recommendation Model Using Open-Source LLM

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Introduction

The e-commerce market is becoming increasingly competitive, requiring advanced recommendation systems. Large marketplaces use sophisticated AI models, but independent online store operators lack such solutions. This project aims to develop an AI-powered product recommendation system tailored for self-managed online stores. The objectives of the project are as follows:

- Develop an AI-based search and recommendation system that analyzes user input sentences and recommends the TOP 5 most relevant products based on semantic similarity.
- Utilize NLP, vector search technology, RAG with LLM for refined search experience.

Method

Data & Model

- OpenAI Embeddings + Qdrant for semantic search
- LLaMA 3.1 8B Instant setup for on-premise inference
- LangChain for Retrieval-Augmented Generation (RAG)

System Development

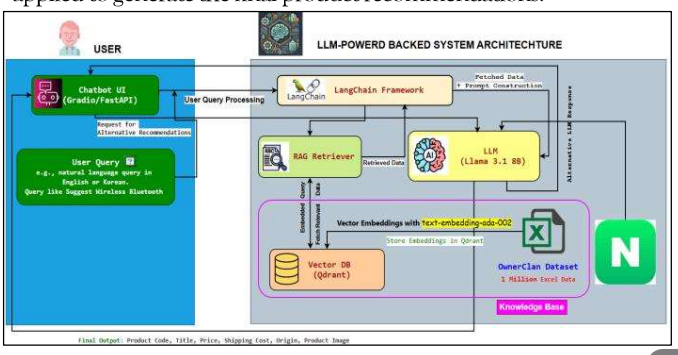
- FastAPI backend, Gradio frontend
- Chatbot GUI with image support
- Multilingual (EN/KO/JA/TH) and modifiable field support

Features

- Fallback (Alternate) recommendation logic.
- User Auth: Login/register, chat history, voice input.
- Dedicated Naver product search: External product lookup.
- Evaluation dashboard: Recommendation accuracy/response time.

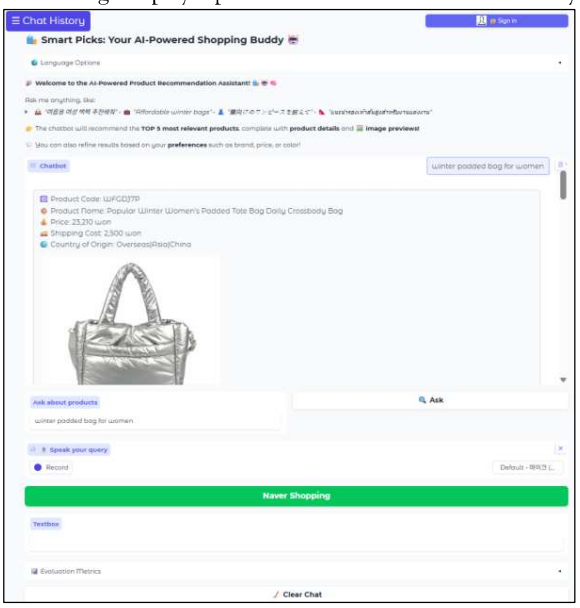
Architecture

The proposed architecture illustrates how user queries are processed, how products are retrieved from both internal (OwnerClan) and external (Naver API) sources, and how LLM-based reasoning is applied to generate the final product recommendations.

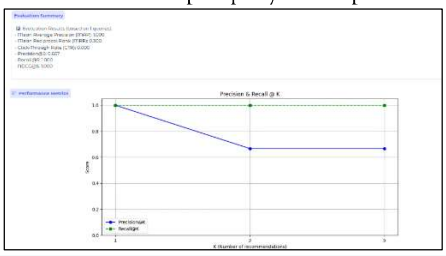


Application & Result

- Successfully processed 1M product dataset (1000 chunks).
- Achieved ≤5s response time for recommendations
- Implemented fallback logic for alternate suggestions.
- Developed complete AI-chatbot interface with:
  - User authentication
  - Chat history
  - Voice input
  - Naver product search integration
  - Image display capabilities with evaluation functionality



- Evaluation metrics per query and response:



Conclusions

- Developed an on-premise AI recommender for independent stores (1M products, ≤5s latency).
- Combines Qdrant vector search with LLaMA reasoning and Naver API fallback.
- Key features: multilingual support, image preview, chat history, voice input, and performance dashboard.
- Future work: Enhanced evaluation metrics (Click-through rate), broader API integration (e.g., Coupang, Amazon).

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